

Investigating Situated Learning Theory through an Augmented Reality Mobile Assistant for Everyday STEM Learning

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ABSTRACT

Situated learning theory (SLT) suggests learning should take place within authentic contexts to be effective. Research in virtual and augmented reality (AR) tend to use SLT to ground their work. Yet, the effectiveness of SLT to ground designs is still questionable. This work investigates situated learning through an AR mobile application called Objectica that seeks to teach STEM (Science, Technology, Engineering, Mathematics) concepts using authentic everyday objects. A between-subjects study compared Objectica with a version not grounded in SLT. Initial results show no significant differences in the effectiveness of the two versions, questioning impact of SLT in educational app design.

Index Terms: Human-Computer Interaction—Interaction Paradigms—Augmented Reality; Education—STEM Learning—Situated Learning Theory

1 INTRODUCTION

The situated learning theory (SLT), as posited by Lave and Wenger [1] views learning as a social process, where learning happens in the learner's context through participation in a community of practice. The theory contrasts from traditional views on decontextualized learning by suggesting that knowledge acquisition is embedded in learner's everyday activity, context, and culture. It thus advocates for educational experiences that are experiential, interactive, and based in the learner's context. In recent years, Augmented Reality (AR) has become a key area of educational research, particularly in the context of situated learning. Its ability to provide immersive, context-rich learning experiences makes it a valuable tool for education. A 2019 survey by Wen and Looi [2] highlighted limited research efforts that explicitly integrate AR with SLT, particularly in science learning. Only two studies in this review ran empirical studies, but both found positive impacts on learning outcomes and motivation. A review by Dunleavy and Dede [3] also found that majority of AR mobile applications grounded in SLT are geared towards K-12 learners. Further, the primary focus of existing literature has been in the domain of language learning, making it pertinent to study the impact of situated learning in STEM education. Thus, it is evident that a gap in knowledge exists when considering the effectiveness of AR-based situated learning experiences in STEM for adult learners.

To that end, we investigated the use of SLT as a grounding for the design of AR learning applications by comparing an AR mobile app designed based on situated learning with one that was not. We hypothesized that embedding learning experiences in a learner's

natural environment would improve the perceived relevance and motivation to engage in STEM learning. To test this hypothesis, we ran a between-subjects study with 43 participants. Below, we describe the AR application followed by study design and preliminary results.

2 OBJECTICA: A STEM LEARNING MOBILE APP

Objectica is an Android application embodying a digital companion whose purpose is to operationalize situated learning to support STEM learning. It leverages the inherent scientific value objects in a learner's everyday environment possess. We designed two distinct versions of this app for the purpose of our research.

2.1 Objectica AR: Grounded in Situated Learning Theory

In Objectica AR, a user starts by logging in and selecting a virtual avatar to act as their digital companion. The selected digital companion serves as an embodiment of social interaction within the app. The app periodically prompts the user to engage with their environment using their smartphone's camera. When a user launches the camera app, Objectica AR employs You Only Look Once (YOLO) [4], a real-time object-detection algorithm to find scientifically interesting objects in the frame. A list of objects and their associated science facts is stored in a connected Google Firebase database. When an object detected in a frame matches one present in the database, the object is overlaid with a relevant science fact presented by the chosen digital companion and show in the app. The user can then read the fact, rate how interesting it was to them, and bookmark it if they find it interesting. As a learner discovers new objects and STEM facts, these are saved in Objectica AR for later access.

2.2 Objectica non-AR: Decontextualized App

We also developed Objectica non-AR, a decontextualized version for the control condition. This version does not use a learner's camera app. Instead, learners are periodically sent science facts about random objects as push notifications from the same Firebase database used by Objectica AR. This takes away the situated aspect of learning but keeps consistent the learning content that is shown in both apps. Users still choose a digital companion, access discovered science facts later, and have the ability to bookmark them.

3 STUDY DESIGN AND DATA COLLECTION

To assess the impact situated learning has on perceived relevance and motivation, we ran a between-subjects study with a control and an experimental group. Participants, all 18 years or older with access to an Android smartphone were recruited from the university's recruitment platform. We ran a total of 43 participants (14 female, 28 male, 1 non-binary; average age: 22.81 years; age range: 18 to 29 years). Most participants (39) came from a computer science background while a few belonged to digital arts (2) and sociology (2). Of these, 26 participants had prior experience with STEM learning apps while 17 did not. Before the start of the study, participants filled out a consent form and a demographic questionnaire. At the scheduled time, participants joined a Zoom session where a researcher onboarded them onto the study. Participants were randomly assigned to one

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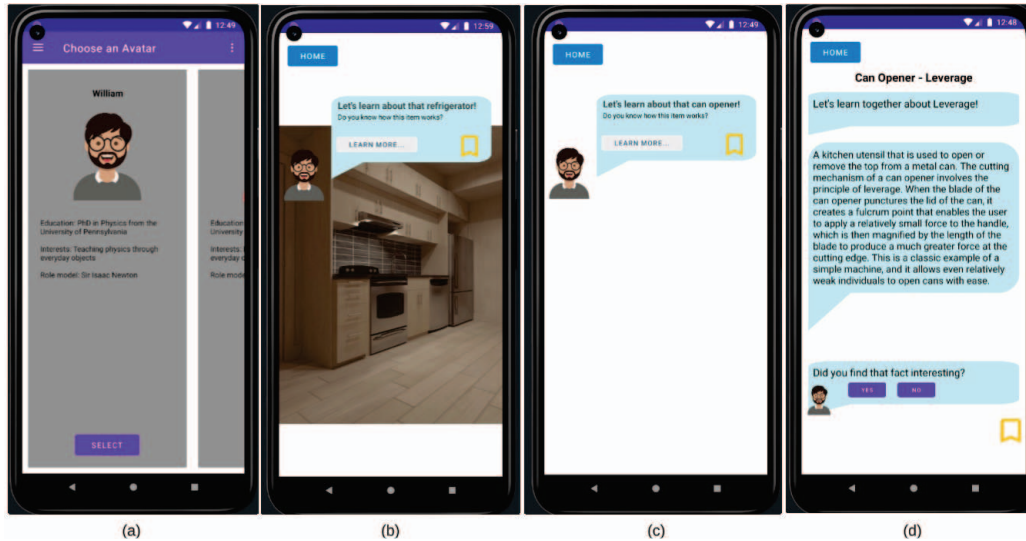


Figure 1: (a) Digital Companion Selection; (b) Science Fact in Objectica AR; (c) and (d) Science fact in Objectica non-AR

Table 1: Means of self-report measures

Condition	SUS	Int/Enj	Eff/Imp	Val/Use	SI-Motivation
O-AR	63.43	4.51	3.97	4.42	3.00
O-non-AR	62.36	4.18	3.64	4.24	2.55

of the two conditions. The experimental group was asked to install Objectica AR while the control group installed Objectica non-AR. All participants were given a walkthrough of how to use their app, as elaborated on in sections 2.1 and 2.2 and were encouraged to ask any clarifying questions. Throughout the week-long study, both apps sent push notifications every six hours to prompt engagement, though participants were free to use Objectica AR as frequently as they wished. The study concluded at the end of the 7th day with participants completing a questionnaire.

The post-study questionnaire was designed to evaluate participant motivation, relevance to learning content, and usability of both apps. It incorporated three sub-scales of the intrinsic motivation inventory (IMI); interest/enjoyment, effort/importance, and value/usefulness to assess motivation. The relevance sub-scale from the ARCS model was used to gauge relevance of the learning content to participants. It also administered the System Usability Scale (SUS) to assess the comparative usability between the two apps. Additionally, two single items (SIs), one each for relevance and motivation were used.

4 PRELIMINARY RESULTS

Based on results from Shapiro-Wilks test for normality, we used an independent samples t-test for normally distributed data and a Mann-Whitney U test for others. The tests showed no significant differences in means between the two groups for self-report measures including SUS, IMI sub-scales, ARCS, and the two single items. However, the mean scores for Objectica AR were consistently higher across all self-report measures, except for ARCS and the relevance single item as seen in table 1. We also looked at whether gender played a role in perceived relevance or motivation. Interestingly, male participants found significantly higher ($t(40) = .441, p = 0.044$) value/usefulness in both Objectica AR and non-AR as compared to females. We also found that participants who had prior experience of using apps that supported STEM learning reported higher SUS

scores than others, although a correlation test yielded non-significant results. Finally, a high positive correlation ($r(42) = .799, p = <0.001$) was found between IMI's interest/enjoyment and value/usefulness sub-scales for both conditions.

5 DISCUSSION AND CONCLUSION

Our preliminary analysis, while not yielding statistically significant differences, did reveal a consistent pattern of higher average self-report scores for Objectica AR, grounded in SLT, in most measures of relevance and motivation. This observed trend suggests a potential positive impact of Objectica AR, aligning with the principles of SLT. The absence of significant findings might be attributed to the relatively small sample size in our between-subjects study design, which could limit the statistical power necessary to detect subtle differences. In response, we are planning to expand our participant pool. Additionally, our results highlight the importance of personal factors, such as prior experience with STEM learning apps and gender, in influencing one's perception of STEM learning. This insight supports the notion that individual differences can play a crucial role in the effectiveness of educational technologies. In conclusion, while our study did not find statistically significant differences, our findings indicate a neutral to potentially positive influence on relevance and motivation in STEM learning when supported by AR grounded in SLT, warranting continued research in this area.

ACKNOWLEDGMENTS

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